

Power Markets

Trading and Hedging a Retail Electricity Portfolio



Build the portfolio and HPFC, design the hedge, manage delivery and advise the trading desk.

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Disclaimer: This simplified educational exercise includes constructed scenario assumptions. It does not represent an executable historical trading strategy or financial, trading, or risk-management advice.

Variable glossary

The notation below is used throughout the project. Unless stated otherwise, interval energy volumes refer to one quarter-hour.

Symbol(s)	Meaning	Unit
p, t, τ, j, b	customer profile, delivery interval, summation interval, futures product, and non-overlapping optimization block	index
E_p, n_p, e_p	annual portfolio energy for profile p , customer count, and annual energy per customer	MWh, count, MWh
$s_{p,t}, L_{p,t}, L_t$	normalized load-profile value, forecast load for profile p , and total forecast load	profile unit, MWh, MWh
$P_{base}, P_{peak}, P_{off}, P_{slice}$	Base, Peak, implied Off-Peak, and applicable slice price	EUR/MWh
$N_{all}, N_{peak}, N_{off}$	number of all, Peak, and Off-Peak quarter-hours in a delivery period	intervals
$s_t, \text{mean}_{slice}(s), \text{HPFC}_t$	historical price-shape factor, its mean within a slice, and the hourly price forward curve value	factor, factor, EUR/MWh
$\Delta t, x_f, D_{j,t}, H_t$	interval duration, futures position, delivery indicator, and hedge energy	hours, MW, 0 or 1, MWh
R_t^{DA}, P_t^{DA}	residual forecast volume settled Day-Ahead and realised Day-Ahead price	MWh, EUR/MWh
A_t, I_t, P_t^{imb}	actual load, imbalance volume, and realised imbalance price	MWh, MWh, EUR/MWh
$C_{futures}, C_{forecast}$	fixed futures cost and total cost of procuring the forecast position	EUR
$C_{residual}^{DA}, C_{imb}, C_{final}$	Day-Ahead residual cost, imbalance settlement, and final procurement cost	EUR
$\Pi_{futures}, \text{Premium}_{imb}, \text{Saving}$	financial futures payoff, imbalance premium, and saving relative to the unhedged benchmark	EUR

1. The management decision

You have completed your first review of the customer and market data. Your manager now asks you to recommend a procurement strategy for the company's 2024 customer portfolio using the futures products available on 29 September 2023.

The recommendation should protect the company against adverse wholesale-price movements while keeping the hedge reasonably close to the shape of the expected customer load. It should be evaluated as a risk-management decision, not as a speculative trade.

Standard Base and Peak futures can fix the price of broad delivery blocks, but the customer load is neither flat nor perfectly known. The company therefore faces:

- **shape risk**, because the futures do not exactly match the forecast load; and
- **volume risk**, because actual customer consumption can differ from the forecast.

The procurement process is represented in three stages:

Futures hedge
covers part of the forecast load at agreed futures prices

Day-Ahead shaping
buys or sells the forecast volume not matched by the futures

Imbalance settlement
settles actual consumption minus the Day-Ahead schedule

Your work should show how the cost and risk change as each stage is added.

2. Strategies to evaluate

Compare three strategies:

Strategy	Futures universe
UNHEDGED	No futures
COARSE_CAL	Eligible annual Base and Peak products
GRANULAR	The finest non-overlapping combination of eligible Month and Quarter Base and Peak products

Selecting the futures is part of the task. Use the `market_activity` criterion developed in your data briefing, justify which products are tradable, and demonstrate that each hedge covers 2024 without overlapping delivery periods.

Futures positions are continuous MW and must be non-negative. Transaction costs, bid-ask spreads, and integer contract lots are not modelled.

3. Stage 1 - Forecast the customer portfolio

Inputs: the customer table in the Data Description and `slp_profiles.csv`

Construct the quarter-hourly forecast load of each customer group and the total portfolio.

For profile p , let E_p be annual portfolio energy, n_p the number of customers, and e_p annual consumption per customer:

$$E_p = n_p * e_p$$

Let $s_{p,t}$ denote the supplied normalized profile value at quarter-hour t . Scale the corresponding standard load profile:

$$L_{p,t} = E_p * s_{p,t} / (\sum_{\tau} s_{p,\tau})$$

Then calculate:

$$L_t = \sum_p L_{p,t}$$

Reconcile the resulting annual energy with the customer table and distinguish correctly between quarter-hourly MWh and average MW.

Explain how the customer mix affects the daily and seasonal load shape and the potential use of Base and Peak futures.

4. Stage 2 - Construct the HPFC

Inputs: `futures_prices.csv` and `shape_factors.csv`

Your manager needs an hourly valuation of the portfolio, but the futures market provides only average prices for broad delivery blocks. Construct a simplified Hourly Price Forward Curve (HPFC), represented on the project's quarter-hourly grid, using the eligible non-overlapping products selected for the granular strategy.

The HPFC combines two sources of information:

- traded futures determine market-consistent price levels for overlapping delivery blocks; and
- an hourly model determines how prices vary within these blocks.

The following subsections describe these two stages and their reconciliation. Further details and alternative approaches will be discussed in the lectures.

4.1 Infer the Off-Peak price

For each selected delivery period, the Base future gives the average price across all intervals, while the Peak future gives the average across only the Peak intervals. To shape Peak and Off-Peak hours separately without changing the quoted Base value, the prices must satisfy:

$$P_{base} * N_{all} = P_{peak} * N_{peak} + P_{off} * N_{off}$$

Rearranging this identity gives the implied Off-Peak price:

$$P_{off} = (P_{base} * N_{all} - P_{peak} * N_{peak}) / N_{off}$$

where N_{all} , N_{peak} , and N_{off} are the numbers of all, Peak, and Off-Peak quarter-hours in the applicable delivery period.

4.2 Apply the historical price shape

Within Peak and Off-Peak separately, write s_t for the supplied historical shape factor and rescale it:

$$HPFC_t = P_{slice} * s_t / \text{mean}_{slice}(s)$$

P_{slice} is either the quoted Peak price or the implied Off-Peak price.

4.3 Reconcile the curve

For every calibration period, verify that:

- the mean HPFC across Peak intervals equals the quoted Peak future; and
- the mean HPFC across all intervals equals the quoted Base future.

Use eligible products that were not used to build the HPFC as independent reconciliation checks. This tests whether the curve is also consistent with coarser or overlapping market information. Small differences caused by quotes rounded to cents are acceptable.

Conclude this stage by explaining why a contract-consistent HPFC is useful for valuation but is not a forecast of realised Day-Ahead prices.

5. Stage 3 - Optimize the futures hedge

Electricity retailers seek to manage their exposure to both price and volume risk. This section outlines basic futures-hedging principles for the forecast portfolio; the remaining realised volume risk is addressed during delivery. Further details and alternative hedging approaches will be discussed in the lectures.

Inputs: the forecast portfolio, completed HPFC, and `futures_prices.csv`

Let x_j be the MW position in futures product j . Construct the energy represented by the hedge in every quarter-hour:

$$H_t = \Delta t * \sum_j x_j * D_{j,t} \text{ with } \Delta t = 0.25 \text{ h}$$

A Base product delivers in every interval of its delivery period. A Peak product delivers only in the defined Peak intervals.

5.1 Economic constraint - value neutrality

Within each non-overlapping optimization period b , require:

$$\sum_{t \in b} H_t * HPFC_t = \sum_{t \in b} L_t * HPFC_t$$

Apply the constraint to each non-overlapping delivery block in the selected strategy. The coarse hedge therefore has one annual value constraint, while the granular hedge has one constraint for each selected Month or Quarter block.

5.2 Minimize the shape mismatch

Among the positions satisfying value neutrality, solve:

$$\min_x \sum_t (L_t - H_t)^2$$

subject to:

$$x_j \geq 0$$

5.3 Check the hedge

For each strategy, report:

- futures positions in MW;
- hedge notional energy in MWh;
- hedge and forecast values under the HPFC;
- value-neutrality error;
- residual-load RMSE; and
- absolute residual Day-Ahead volume.

Explain why value neutrality does not necessarily produce volume neutrality, and assess which price and shape risks remain after optimization.

6. Stage 4 - Shape the forecast position Day-Ahead

Input: day_ahead_prices.csv

The futures are financial contracts and do not change the physical customer schedule. The company schedules its forecast load Day-Ahead, while the part not covered economically by the futures remains exposed to the Day-Ahead price:

$$R_t^{DA} = L_t - H_t$$

Calculate the cost of the forecast position:

$$C_{forecast} = C_{futures} + \sum_t R_t^{DA} * P_t^{DA}$$

Validate the result through the equivalent financial representation:

$$C_{forecast} = \sum_t L_t * P_t^{DA} - \Pi_{futures}$$

For UNHEDGED, hedge_energy[t] is zero and the complete forecast load remains exposed to Day-Ahead prices.

Reconcile the two cost representations. Interpret both signs of day_ahead_residual: what action is required when it is positive, and how should a negative value be handled?

7. Stage 5 - Settle the volume deviation

Inputs: actual_portfolio_load.csv and imbalance_prices.csv

During delivery, actual consumption differs from the Day-Ahead schedule. Calculate:

$$I_t = A_t - L_t$$

The imbalance settlement is:

$$C_{imb} = \sum_t I_t * P_t^{imb}$$

Final procurement cost becomes:

$$C_{final} = C_{futures} + C_{residual}^{DA} + C_{imb}$$

Also isolate the price consequence of being imbalanced:

$$Premium_{imb} = \sum_t I_t * (P_t^{imb} - P_t^{DA})$$

Interpret positive and negative imbalance_volume and the corresponding settlement cashflow. The Day-Ahead schedule is the same forecast portfolio under all three strategies; show what this implies for imbalance settlement and explain which risk the futures hedge does not remove.

8. Stage 6 - Advise management

Compare the strategies using at least:

Measure	Purpose
Final annual procurement cost	Total economic outcome
Final average price in EUR/MWh of actual load	Comparable unit cost
Saving versus unhedged	Procurement P&L benchmark
Monthly average procurement-price standard deviation	Descriptive cost stability
Monthly price range	Exposure to seasonal extremes
Absolute residual Day-Ahead volume	Remaining spot exposure
Residual-load RMSE	Hedge shape quality
Imbalance settlement and premium	Cost of forecast error

Define procurement P&L relative to the unhedged benchmark:

$$Saving = C_{final}^{unhedged} - C_{final}^{strategy}$$

A positive value indicates a lower realised procurement cost. It is a backtest result, not evidence that the hedge was designed as a speculative profit strategy.

Recommend one strategy to management. Support the recommendation with both cost and risk evidence; the strategy with the lowest realised annual cost is not automatically the best hedge.

9. Stage 7 - Interpret a severe price event

Identify a relevant high-price day or interval in the 2024 Day-Ahead data.

Use the selected event and the results already produced to show the Day-Ahead price, customer load, hedge, and residual exposure. Compare how the three strategies were exposed and explain how the event affected procurement cost.

This is primarily an interpretation exercise, not a new optimization or simulation task. A simple sensitivity showing what would happen if the high prices were larger or lasted longer may be used to support the discussion, but it is not required.

Conclude the event analysis by answering: **If the company had not hedged, how could an extreme spot-price event affect its liquidity and ability to continue serving customers? Which risks would remain even with the hedge?**

Avoid assuming that one expensive day automatically causes bankruptcy. Base the conclusion on the size and duration of the exposure and the protection provided by each strategy.

10. Optional extensions

Price the customer product

Your manager now asks what fixed energy price the company should offer its customers. Develop a transparent pricing approach using only information that would have been available when the offer was made.

Decide which cost and risk components should enter the price, state any additional assumptions or information you require, and explain how the selected hedge affects the result. Clearly separate an ex-ante price from conclusions that rely on realised 2024 data.

Test another hedge design

As a separate extension, you may:

- construct a QUARTER_ONLY hedge using Q1-Q4 Base and Peak;
- propose another non-overlapping product set; or
- add one realistic trading constraint and discuss its effect.

11. Deliverables and evidence

Each team must submit:

1. a reproducible notebook, script, or workbook; and
2. a 10-15 minute presentation communicating the method, evidence, and management recommendation.

The analysis and presentation must include:

- an HPFC reconciliation table;
- a futures-position table;
- a final cost-and-risk comparison table;
- the customer portfolio shape and composition;
- the HPFC shape and contract-reconciliation evidence;
- the forecast load, hedge shape, and residual exposure;
- a monthly procurement cost or price comparison; and
- the selected severe-price-event analysis.

Teams are free to choose the presentation structure, number of slides, chart types, and any additional evidence.

12. Assessment guidance

Area	Weight
Data preparation and portfolio validation	20%
HPFC and hedge-optimization correctness	25%
Day-Ahead and imbalance accounting	20%
Economic interpretation and risk understanding	20%
Communication and independent insight	15%

Strong submissions will reproduce the required accounting and optimization correctly, identify limitations without losing sight of the simplified model, and make a clear commercial recommendation supported by evidence.

13. Assumptions and limitations

Return to the assumptions identified in your initial data briefing. Select at least three and explain how they could affect the hedge, the procurement result, or the management recommendation. Where possible, describe the likely direction of the effect.